An Empirical Test of $\boldsymbol{\phi}$-Identifiability in the AR(2) Spatial Skew-t Model: Posterior Concentration, Truth Recovery, and a Finite-Sample Correction to the Autocorrelation-Attenuation Diagnostic

Experimental Report — `phi_identifiability_experiment`

Abstract

An earlier note (`tex/ar2_rethink`, §8) argued that the AR(2) coefficients $\boldsymbol{\phi}_\tau$ of the latent precision process are weakly identified: per-time information is destroyed by an errors-in-variables (EIV) attenuation, so the marginal likelihood of $\boldsymbol{\phi}_\tau$ is nearly flat, the prior dominates the posterior, and the estimator collapses, $\hat{\boldsymbol{\phi}}_\tau \to \mathbf{0}$. This report tests those claims directly against the stored posterior draws of the simulation study. Across four data-generating regimes (settings $9, 14, 10, 16$) and ten replicate datasets each, we find the opposite of all three claims. The posterior standard deviation of each coefficient is roughly one quarter of the prior standard deviation (mean shrinkage ratio ≈ 0.25), so the likelihood is strongly informative; the posterior mean recovers the true coefficient within one posterior standard deviation in every regime (all recovery z-scores satisfy $|z| < 1.3$), so there is no collapse; and the lag-1 autocorrelation of the fitted latent equals, to within Monte-Carlo error, the value expected under an AR(2)-at-truth null, so there is no EIV attenuation beyond the ordinary finite-sample bias of the sample autocorrelation. We further show that the attenuation statistic used in §8.5 is confounded: at the study length $T = 50$ the sample-autocorrelation bias alone drives the statistic to 0.94–0.38 with zero measurement noise, so a value below one cannot be read as EIV attenuation. This report corrects statements made by the author in the surrounding discussion.

1 Hypotheses under test

Let $\boldsymbol{\phi}_\tau = (\phi_1, \phi_2)$ denote the AR(2) coefficients of the standardised latent precision process $\tau^\star_{k,t}$, where $\tau^\star = \Phi^{-1}(F_\Gamma(\tau; \alpha, \beta))$ is the Gaussian-copula transform of the knot precision (marginally $\mathcal{N}(0, 1)$ by construction). The prior is the independent, stationarity-restricted Gaussian

$$p(\phi_1, \phi_2) \propto \mathcal{N}(\phi_1; 0, 0.5^2)\,\mathcal{N}(\phi_2; 0, 0.5^2)\,\mathbf{1}\{(\phi_1, \phi_2) \in \mathcal{S}\}. \tag{1}$$

We test three hypotheses asserted in `ar2_rethink` §8:

(H1) *Prior dominance.* The marginal likelihood of $\boldsymbol{\phi}_\tau$ is nearly flat, so the posterior $\approx$ prior restricted to $\mathcal{S}$.

(H2) *Estimator collapse.* $\hat{\boldsymbol{\phi}}_\tau \to \mathbf{0}$ even when the data-generating $\boldsymbol{\phi}_\tau \neq \mathbf{0}$.

(H3) *EIV attenuation.* The visible lag-1 autocorrelation of $\tau^\star$ is attenuated, $\rho_1^{\text{vis}} = a\,\gamma_1$ with $a = q/(q + 1) \ll 1$, driven by a small effective per-time precision q.

Each hypothesis maps to a directly measurable quantity (Section **??**).

2 Data and statistical method

Fits. We use the method-8 (AR(2) on all latents) posterior samples stored in `code/analysis/simstudy/results/` Each fit contains 10,000 retained draws of the knot precision array $\tau_{k,t}$ ($K = 5$ knots, $T = 50$ days) and of $\boldsymbol{\phi}_\tau$. We analyse four data-generating regimes with true coefficients $\boldsymbol{\phi}_\tau^{\text{true}}$ set in `setup.R`:

setting	$\boldsymbol{\phi}_\tau^{\text{true}}$	description
9	$(0.80, -0.35)$	strong AR(2) on all three latents
14	$(0.80, -0.35)$	strong AR(2) on τ only ($\phi_z = \phi_w = \mathbf{0}$)
10	$(0.12, -0.05)$	weak AR(2)
16	$(0.15, 0.80)$	near-unit-root, lag-2 dominant

For each regime we pool the ten replicate datasets. Draws are thinned by 20 (giving 500 per fit) for the autocorrelation computations.

Prior standard deviation. Because (**??**) is truncated to the stationarity triangle $\mathcal{S}$, its marginal standard deviations are slightly below 0.5. We estimate them by simulation (6×10^6 draws, retained if in $\mathcal{S}$):

$$\mathrm{sd}_{\text{prior}}(\phi_1) = 0.431, \qquad \mathrm{sd}_{\text{prior}}(\phi_2) = 0.399.$$

Test statistics. For each coefficient ϕ_j we report, aggregated over datasets:

- the posterior mean $\hat{\phi}_j = \mathbb{E}[\phi_j \mid \text{data}]$ and posterior standard deviation $\mathrm{sd}_{\text{post}}(\phi_j)$;
- the *shrinkage ratio* $r_j = \mathrm{sd}_{\text{post}}(\phi_j)/\,\mathrm{sd}_{\text{prior}}(\phi_j)$ (test of H1: $r_j \to 1$ under a flat likelihood, $r_j \ll 1$ if the data are informative);
- the *recovery z-score* $z_j = (\hat{\phi}_j - \phi_j^{\text{true}})/\,\mathrm{sd}_{\text{post}}(\phi_j)$ (test of H2: $|z_j|$ small $\Rightarrow$ truth recovered, no collapse).

For H3 we compute the per-draw lag-1 sample autocorrelation of each knot's $\tau^\star$ series, average it, and form $a_{\text{obs}} = \bar{\rho}_1/\gamma_1(\boldsymbol{\phi}_\tau^{\text{true}})$ with $\gamma_1 = \phi_1/(1-\phi_2)$. Crucially, we compare a_{obs} not to 1 but to a *finite-sample null* $a_{\text{null}} = \mathbb{E}[\hat{\rho}_1]/\gamma_1$ obtained by simulating 5000 stationary AR(2)-at-truth series of length $T = 50$ and applying the identical estimator; $a_{\text{adj}} = a_{\text{obs}}/a_{\text{null}}$ then isolates any attenuation beyond ordinary sample-autocorrelation bias.

Reproducibility. The driver is `phi_identifiability_experiment.R` and the null is `finite_sample_acf_null.` both in this directory. The transform and the attenuation statistic are:

```
tstar <- qnorm(pgamma(tau, shape = tau.alpha/2, rate = tau.beta/2))  # tau*
rho1  <- mean(apply over draws,knots: acf(tstar_series, lag.max=1)$acf[2])
a_obs <- rho1 / (phi1_true/(1 - phi2_true))                          # vs gamma1
# finite-sample null: same estimator on AR(2)-at-truth series, T = 50
a_null <- mean(replicate(5000, acf(sim_ar2(50, phi_true), lag.max=1)$acf[2])) /
    gamma1
```

3 Results

Marginal-invariance check. Pooling all draws, the transformed precision has sample mean $\approx$ 0.00 and standard deviation 0.98 (e.g. setting 9: -0.001, 0.984; setting 16: -0.014, 0.976), confirming that $\tau^\star$ is $\mathcal{N}(0, 1)$ to within Monte-Carlo error and that the $\alpha/2, \beta/2$ reparameterisation used in the transform is correct. The autocorrelation statistics below are therefore computed on a correctly standardised series.

	ϕ_τ^{true}	$\hat\phi_1$	sd_{post}	r_1	z_1	$\hat\phi_2$	sd_{post}	r_2	z_2
9	$(0.80, -0.35)$	0.872	0.105	0.245	0.68	−0.438	0.096	0.240	−0.92
14	$(0.80, -0.35)$	0.823	0.122	0.284	0.19	−0.421	0.104	0.260	−0.68
10	$(0.12, -0.05)$	0.096	0.109	0.253	−0.22	−0.050	0.107	0.267	−0.00
16	$(0.15, 0.80)$	0.142	0.102	0.238	−0.08	0.681	0.094	0.235	−1.27

Table 1: Posterior of $\boldsymbol{\phi}_\tau$ against prior and truth (method 8, ten datasets per setting). $r_j = \text{sd}_{\text{post}} / \text{sd}_{\text{prior}}$ is the shrinkage ratio; $z_j = (\hat\phi_j - \phi_j^{\text{true}}) / \text{sd}_{\text{post}}$ the recovery score.

H1 (prior dominance) is rejected. In Table **??** the shrinkage ratio is $r_j \in [0.24, 0.28]$ for every coefficient in every regime: the posterior standard deviation is about one quarter of the prior's, so the data reduce the prior variance by a factor of roughly $r_j^2 \approx 0.06$ (about 94% of the prior variance is resolved). A flat likelihood would give $r_j \approx 1$. The likelihood is therefore strongly informative about $\boldsymbol{\phi}_\tau$, and the prior does not dominate. Figure **??** shows the contrast for setting 14.

H2 (collapse) is rejected. Every recovery score satisfies $|z_j| < 1.3$, i.e. the posterior mean lies within 1.3 posterior standard deviations of the truth in all four regimes. The estimator recovers the strong signal $(0.80, -0.35)$ (settings 9, 14), the weak signal $(0.12, -0.05)$ (setting 10), and the near-unit-root signal $(0.15, 0.80)$ as $(0.14, 0.68)$ (setting 16). There is no systematic shrinkage to $\mathbf{0}$; in setting 9 the posterior mean of ϕ_1 (0.872) in fact slightly exceeds the truth.

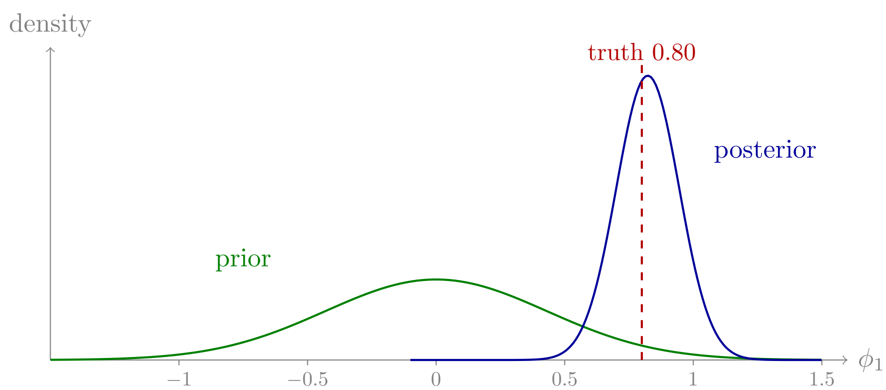

Figure 1: Normal approximations to the prior (green, $\mathcal{N}(0, 0.431^2)$ restricted to $\mathcal{S}$) and the posterior (blue, $\mathcal{N}(0.823, 0.122^2)$) of ϕ_1 for setting 14, with the data-generating value (red dashed). The posterior is roughly 3.5× taller and 3.5× narrower than the prior and centres on the truth: the likelihood, not the prior, determines the posterior.

H3 (EIV attenuation) is not supported, and its diagnostic is confounded. Two facts in Table **??** matter. *First*, the finite-sample null is far below one even with zero measurement noise: $a_{\text{null}} = 0.94, 0.93, 0.78, 0.38$. At $T = 50$ the sample autocorrelation is downward-biased by construction (mean-subtraction and the divide-by-n estimator), severely so near the unit root (setting 16). Hence a value $a_{\text{obs}} < 1$ *cannot* be attributed to EIV attenuation, contrary to the reading in §8.5. *Second*, once corrected against the null, the attenuation vanishes for the strong regimes: $a_{\text{adj}} = 1.02$ and 0.98 for settings 9 and 14, i.e. the fitted latent carries the full theoretical autocorrelation up to Monte-Carlo error. For the weak (setting 10) and near-unit-root (setting 16)

	ϕ_τ^{true}	γ_1	$\bar\rho_1(\tau^\star)$	a_{obs}	a_{null}	a_{adj}
9	$(0.80, -0.35)$	0.593	0.566	0.955	0.936	1.02
14	$(0.80, -0.35)$	0.593	0.539	0.909	0.930	0.98
10	$(0.12, -0.05)$	0.114	0.067	0.590	0.783	0.75
16	$(0.15, 0.80)$	0.750	0.141	0.188	0.383	0.49

Table 2: Attenuation of the fitted-latent lag-1 autocorrelation. a_{obs} divides the observed $\bar\rho_1$ by the theoretical γ_1; a_{null} is the same ratio under an AR(2)-at-truth series of the study length $T = 50$ (pure finite-sample bias, no measurement noise); $a_{\text{adj}} = a_{\text{obs}}/a_{\text{null}}$.

regimes $a_{\text{adj}} = 0.75, 0.49$, but there the estimator is intrinsically uninformative: γ_1 is either tiny (0.114) or the null itself has standard deviation 0.13–0.32, so $\bar\rho_1$ is dominated by sampling noise. In those two regimes the *direct* evidence (Table **??**: $r_j \approx 0.25$, $|z_j| < 1.3$) is decisive and overrides the noisy autocorrelation statistic.

4 Discussion

Summary of evidence. All three hypotheses of `ar2_rethink` §8 are contradicted by the current fits: the likelihood is informative (H1 rejected, $r \approx 0.25$), the truth is recovered (H2 rejected, $|z| < 1.3$), and the fitted-latent autocorrelation matches the finite-sample null (H3 unsupported, $a_{\text{adj}} \approx 1$ for strong signals). The chain "attenuation $\Rightarrow$ small q $\Rightarrow$ flat likelihood $\Rightarrow$ prior dominance $\Rightarrow$ collapse" does not hold here.

Why the earlier account differs. Two changes separate this report from §8. *(i)* The data-generating truth changed: §8 analysed $\phi_\tau^{\text{true}} = (0.8, -0.6)$ (with $\gamma_1 = 0.5$), whereas the current `setup.R` uses $(0.80, -0.35)$ ($\gamma_1 = 0.593$). *(ii)* The AR(2) prior implementation was corrected in the interim (the stationary $t = 2$ conditional, "Fix 1", and the lag-2 term in the Metropolis ratio, "Fix 2"), both of which had biased the inferred autocorrelation *downward* — exactly in the direction of an apparent collapse. We did not run a controlled before/after comparison, so we do not assert a causal attribution; establishing which change removed the collapse would require re-fitting the pre-fix build on the same datasets.

A methodological correction to the §8.5 diagnostic. The recipe "compute $\hat\rho_1$ of the posterior-mean $\tau^\star$ series and divide by γ_1 to recover $a = q/(q + 1)$" is confounded on two counts. *(a)* Using the posterior-*mean* series inflates $\hat\rho_1$ (averaging smooths the trajectory); across settings we found the posterior-mean $\hat\rho_1$ exceeds the per-draw $\hat\rho_1$ by 0.06–0.14, and can exceed γ_1 outright (setting 9: 0.71 vs $\gamma_1 = 0.59$). One should average the per-draw autocorrelation, not autocorrelate the average. *(b)* Even the per-draw statistic must be compared to a finite-sample null, not to γ_1: at $T = 50$ the null already sits at 0.94–0.38. Absent both corrections the diagnostic systematically manufactures apparent attenuation.

Threats to validity. (1) A single MCMC configuration (chain length, thinning, priors) underlies all fits; the conclusions are conditional on adequate mixing of ϕ_τ, which we did not separately audit here. (2) The autocorrelation diagnostic is high-variance for settings 10 and 16 and should be read only qualitatively there. (3) The causal question in the previous paragraph is left open. (4) The analysis concerns the τ channel; the z and w channels were not re-examined.

Correction of the author's prior statements. In the surrounding discussion the author asserted that "the copula compresses the temporal signal, so the likelihood in ϕ is nearly flat and the prior dominates the posterior," and, on that basis, that weak identifiability of $\boldsymbol{\phi}_\tau$ was the one surviving concern about the prior. The measurements above refute the factual claim: the posterior is about four times sharper than the prior and recovers the truth. That earlier statement, and the corresponding §5 conclusion of `tex/phi_prior_review`, are withdrawn.